Certified Personal Trainer

DOMAIN THREE

# INTAKE AND ONGOING EVALUATION

EXAM WEIGHT 15%

ISSA INTERNATIONAL SPORTS SCIENCES ASSOCIATION

# Assessments - Chapter 7

In practice, the assessment is often the very first time a client meets with a trainer. This is especially true in fitness facilities that offer an assessment as a free membership perk and, therefore, part of orientation. Assessments help a personal trainer create a customized program for their client. It also provides the trainer and client baseline measurements to track progress as they move their program.

In any case, the assessment is the best time to obtain all necessary client information to ensure a safe and effective program. This includes getting medical clearance from their physician and identifying the parameters the fitness program should work within.

# Gathering Data To Customize Programming - Chapter 7

Before designing a fitness program, a fitness professional should collect and review health information and conduct the appropriate assessments.

- Types of assessments
  - Objective assessments: fitness assessments that collect repeatable, measurable data such as body composition or circumference measurement
  - Subjective assessments: fitness assessments that require observation or a subjective, opinion-based measure
    - An easy way to remember the difference between objective and subjective:
      - Objective: Root word = object. You are objective whenever you are discussing an object or something concrete that you can touch. The information that makes up your objective statement should also be concrete, like data or measurements.
      - Subjective: Something you can't point to. It is based on experience, opinions, facts, and emotions.
- The following paperwork should be the first forms a client fills out. Collectively, this paperwork is known as the initial interview packet.
  - Client intake form
    - A basic intake form to gather a client or potential client's demographic information and general health history
  - PAR-Q
    - Stands for physical activity readiness questionnaire (PAR-Q)
    - An intake form to assess a client's readiness to begin a physical activity program and assess injury potential
  - Health history questionnaire
    - A detailed client intake form that gathers information on a client's present and past health and medical history
      - Have you experienced any of the following conditions?
        - Chest pain

        - Diabetes
        - High blood pressure
        - Irregular heartbeat
        - Etc.
    - What medications are you currently taking?
    - What is your current level of physical activity?
  - Liability waiver
    - A short form that, when signed by a client, releases a fitness professional and/or their training facility from any liability should the client be injured while working with them
  - Physician's letter of clearance (as applicable)
    - A signed letter from a client's health care provider stating that they are cleared for physical activity and exercise; this letter should also include any restrictions or limitations the client should adhere to
  - Three-day dietary record
    - A fitness and nutrition intake form that allows clients to log their food consumption for three consecutive days to observe their habits
- Contraindications: factors that serve as a reason to withhold training because of the harm that it may cause
  - Unstable coronary heart disease (CHD): a condition where the heart does not get enough blood or oxygen flow
  - Decompensated heart failure: new or worsening signs and symptoms of heart failure
  - Uncontrolled heart arrhythmias: irregular heartbeat that is not controlled
  - Severe pulmonary hypertension: extremely high blood pressure
  - Severe and symptomatic aortic stenosis: narrowing of the aortic valve that causes shortness of breath or fatigue
  - Acute myocarditis, endocarditis, or pericarditis: inflammation of the heart muscle, heart chambers, or the sac that surrounds the heart, respectively
  - Uncontrolled hypertension: high blood pressure that is not under control with medication or dietary interventions
  - Aortic dissection: a tear in the aorta
  - Marfan syndrome: a genetic disorder affecting connective tissue that commonly affects the heart, eyes, blood vessels, and skeleton
  - Active proliferative retinopathy or moderate or worse non-proliferative diabetic retinopathy: damage to the blood vessels of the eyes

# Biometric Data - Chapter 7

- Body composition: the physical makeup of the body considering fat mass and lean mass
  - Norms

| SEX | LOW/ ESSENTIAL | ATHLETIC | FIT | AVERAGE | HIGH/ OVERWEIGHT |
|---|---|---|---|---|---|
| Males | 4–6 percent | 7–10 percent | 11–16 percent | 17–25 percent | 26 percent or higher |
| Females | 10–12 percent | 13–20 percent | 21–24 percent | 25–31 percent | 32 percent or higher |

  - Key terms
    - Lean body mass: the fat-free mass of the body calculated by total weight minus the weight of body fat
    - Chronic disease: a condition lasting a year or more that limits daily activities and/or requires medical ongoing attention
    - Amenorrhea: the absence or cessation of a menstrual cycle in females
    - Total classifications of body fat are essential fat (required for living) and storage fat (accumulation of adipose tissue beneath the skin)
    - Body composition testing is the determination of fat and lean components of the human body through various methods
      - Dual energy X-ray absorptiometry (DEXA)
        - An X-ray scanning test to determine body composition
        - Most accurate
        - Because it scans bone mass, it can also determine whether a client has osteoporosis
          - Osteoporosis: a skeletal condition that results in weak or brittle bones.
      - Skinfold testing
        - Most common in the fitness industry because it's inexpensive, portable, and appropriate in lab and field settings
        - Uses skinfold calipers
        - Testing Sites

| SEVEN-SITE BODY DENSITY METHOD | THREE-SITE BODY DENSITY METHOD | |
|---|---|---|
| Male and Female | Male | Female |
| Triceps<br>Chest/pectoral<br>Midaxillary<br>Subscapular<br>Suprailiac<br>Abdomen<br>Thigh | Chest/pectoral<br>Abdomen<br>Thigh | Triceps<br>Suprailiac<br>Thigh |
| **Seven-site formula for males** | **Seven-site formula for females** | |
| 1.112 – (0.00043499 × sum of skinfolds) + (0.00000055 × [sum of skinfolds2]) – (0.00028826 × age) = body density | 1.097 – (0.00046971 × sum of skinfolds) + (0.00000056 × [sum of skinfolds2]) – (0.00012828 × age) = body density | |
| **Three-site formula for males** | **Three-site formula for females** | |
| 1.10938 – (0.0008267 × sum of skinfolds) + (0.0000016 × [sum of skinfolds2]) – (0.0002574 × age) = body density | 1.0994921 – (0.0009929 × sum of skinfolds) + (0.0000023 × [sum of skinfolds2]) – (0.0001392 × age) = body density | |

    - Once body density has been determined with the appropriate formula, the Brozek equation can be used to estimate bodyfat percentage for both men and women:
        - ([4.570 / body density] – 4.142) × 100 = bodyfat percentage (BF%)
- Bioelectrical impedance analysis (BIA)
    - A method for body composition measurement using a weak electrical current to measure the resistance of body tissues
    - Can be affected by hydration status
- Hydrostatic weighing
    - A tool to measure body composition using water displacement and tissue density
    - Highly accurate
    - Not common in fitness facilities
- Air displacement Plethysmography (ADP)
    - Body weight is measured before the individual enters the chamber.
    - Body volume is determined by measuring the volume of the empty chamber compared to the volume of the chamber with the individual inside.

- Classification of body types
  - Mesomorphic: muscular
  - Ectomorphic: slim and linear
  - Endomorphic: round

- Body mass index (BMI): a predictive health measure of weight divided by height$^2$
  - Although it's quick and easy, it isn't the best assessment for clients with high lean body mass—such as bodybuilders—since they will have an inflated number
  - The following are standards of BMI:
    - below 18.5 = underweight
    - 18.5–24.9 = normal
    - 25–29.9 = overweight
    - 30 or above = obese
- Waist-to-height ratio: an objective assessment to measure cardiometabolic risk
  - Waist-to-height ratio = (waist circumference / height) × 100
    - Waist measurement by sex
      - Males: The tape is placed around the waist at the belly button. The tape should be snug against the body without being so tight that it presses into the body. The measurement is then recorded.
      - Females: The client places their forefinger and middle finger on top of the belly button. The tape is wrapped around the abdomen, resting just on top of the forefinger. The measurement is then recorded.

| MEN | RATING | WOMEN |
|---|---|---|
| <35 | Underweight | <35 |
| 35–43 | Extremely slim | 35–42 |
| 43–46 | Slender and healthy | 42–46 |
| 46–53 | Healthy ideal weight | 46–49 |
| 53–58 | Overweight | 49–54 |
| 58–63 | Seriously overweight | 54–58 |
| >63 | Extremely obese | >58 |

- Girth measurements
  - Also called circumference measurements: the measurement of the circumference of specific body regions
  - Locations
    - Neck
    - Upper arm
    - Forearm
    - Waist
    - Hips
    - Thigh
    - Calf
- Blood pressure: the force of blood pushing against the walls of the arteries during the two phases of the cardiac cycle
  - Systolic: the pressure in blood vessels when your heart beats (ventricular contraction)
  - Diastolic: the pressure in blood vessels when your heart rests (ventricular filling)

| RANGE | READING (MILLIMETERS OF MERCURY) |
|---|---|
| Normal | Less than 120/80 mm Hg |
| Elevated | Systolic between 120 and 129 and diastolic less than 80 mm Hg |
| Stage 1 Hypertension | Systolic between 130 and 139 or diastolic between 80 and 89 mm Hg |
| Stage 2 Hypertension | Systolic at least 140 or diastolic at least 90 mm Hg |

# Posture and Movement Assessments - Chapter 7

Following the initial paperwork and collection of biometric data, trainers should take clients through posture and movement assessments to collect information needed for exercise programming and the prescription of flexibility and recovery techniques for optimal movement. These assessments can also serve as preventative measures to help clients avoid poor movement patterns that may result in injury, discomfort, or pain.

- Posture assessments
  - Static posture: posture when standing still
  - Dynamic movement assessments: observation and critique of movement patterns or exercise form
  - Common postural deviations a trainer may observe:

| MALALIGNMENT | POSSIBLE TIGHT MUSCLES | POSSIBLE WEAK MUSCLES |
|---|---|---|
| Lordosis | Lower back (erectors), hip flexors | Abdominals (especially obliques), hip extensors |
| Flat back | Upper abdominals, hip extensors | Lower back (erectors), hip flexors |
| Swayback | Upper abdominals, hip flexors | Oblique abdominals, hip extensors |
| Kyphosis | Internal oblique, shoulder adductors (pectorals and latissimus), intercostals | Erector spinae of the thoracic spine, scapular adductors (mid and lower trapezius) |
| Forward Head | Cervical extensors, upper trapezius | Neck flexors |

- Dynamic movement assessments
  - Key Terms
    - Muscular imbalance: when the muscle or muscles on one side of the body are stronger, weaker, or more or less active than the corresponding muscle on the other side of the body
    - Lumbopelvic hip complex (LPHC): the musculature of the hip that attaches to the pelvis and lumbar spine and works to stabilize the trunk and lower extremities
    - Kinetic chain: a system of links—or joints—in the body that generate and transfer force from one to the other
    - Kinetic chain checkpoints: the six anatomical locations of predictable movement patterns where movement dysfunctions can be detected
      - Foot and ankle
        - Joint action
          - inversion, eversion, dorsiflexion, and plantarflexion
      - Knee
        - Joint action
          - Flexion and extension
      - Hips
        - Joint action
          - Flexion and extension, abduction and adduction, and internal and external rotation
        - This checkpoint must be bserved carefully because it controls the movement of the lower extremities
      - Spine
        - Lumbar
        - Thoracic
        - Cervical

        - Shoulders
            - Most moveable joint
            - Less stable
        - Head and neck
            - Shoulder girdle
                - Clavicle, scapula, and coracoid bones of the appendicular skeleton
            - Thoracic spine
            - Rib cage
            - LPHC
            - Cervical spine
    - Closed kinetic chain movement: a movement keeping the distal end of the body segment in action fixed
    - Open kinetic chain movement: a movement in which the distal aspect of the body segment in action is free (i.e., not fixed)
    - Ideal posture: optimal body positioning and structural alignment
    - Q angle: the quadriceps angle formed between the quadriceps muscle and the patellar tendon
    - Overactive muscles: muscles that are shortened beyond the ideal length-tension relationship with high neural activation that feels tight
    - Underactive muscles: muscles that are lengthened beyond the ideal length-tension relationship and are, therefore, inhibited and less capable of producing force
    - Muscle synergies: the activation of a group of muscles to generate movement around a particular joint
    - Muscular force couple: two or more muscles generate force in different linear directions at the same time to produce one movement
- Common movement dysfunctions a trainer may observe
    - Lordosis: the excessive inward curve of the lumbar spine
    - Kyphosis: the exaggerated rounding of the thoracic spine
    - Flat back: an excessive lumbar flexion and posterior pelvic tilt
    - Swayback: a posterior tilt with an excessive extension of the lumbar spine that protrudes the buttocks
    - Scoliosis: the sideways curvature of the spine
    - Handedness: the tendency to use one side of the body more naturally than the other
    - Knee valgus: the position of the knee near the midline of the body (i.e., knock knees)
    - Knee varus: the position of the knee away from the midline of the body (i.e., bowlegged)
    - Excessive forward lean (one of the more common movement dysfunctions)
- Common observations during a squat assessment may include:

| MALALIGNMENT | POSSIBLE TIGHT MUSCLES | POSSIBLE WEAK MUSCLES |
|---|---|---|
| Knee valgus | Gluteus maximus<br>Gluteus medius<br>Vastus medialis | Adductors<br>Biceps femoris<br>TFL<br>Vastus lateralis |
| Foot turnout (eversion and external rotation) | Gastrocnemius (medial)<br>Semitendinosus<br>Semimembranosus<br>Sartorius<br>Gracilis | Soleus<br>Gastrocnemius (lateral)<br>Biceps femoris |
| Forward lean | Gluteus maximus<br>Erector spinae<br>Anterior tibialis | Soleus<br>Gastrocnemius<br>Psoas<br>Rectus femoris<br>Rectus abdominis<br>External obliques |
| Lumbar extension | Gluteus maximus<br>Transverse abdominus<br>Hamstring complex<br>Multifidus | Psoas<br>Rectus femoris<br>TFL<br>Erector spinae<br>Latissimus dorsi |
| Heel elevation | Tibialis (anterior) | Soleus |
| Foot pronation (arch collapse) | Tibialis (anterior and posterior)<br>Gluteus medius<br>Gastrocnemius (medial) | Gastrocnemius (lateral)<br>Biceps femoris<br>TFL |

| MALALIGNMENT | POSSIBLE TIGHT MUSCLES | POSSIBLE WEAK MUSCLES |
|---|---|---|
| Lateral weight shift | Gluteus medius (same side of shift)<br>Tibialis (anterior)<br>Adductors (opposite side of shift) | Adductors<br>TFL (same side of shift)<br>Gastrocnemius<br>Soleus<br>Piriformis<br>Gluteus medius<br>Biceps femoris (opposite side of shift) |

▶ Common movement dysfunctions a trainer may observe during an overhead squat assessment

| MALALIGNMENT | POSSIBLE WEAK MUSCLES | POSSIBLE TIGHT MUSCLES |
|---|---|---|
| Knee valgus | Gluteus maximus<br>Gluteus medius<br>Vastus medialis | Adductors<br>Biceps femoris<br>TFL<br>Vastus lateralis |
| Forward lean | Gluteus maximus<br>Erector spinae<br>Anterior tibialis | Soleus<br>Gastrocnemius<br>Psoas<br>Rectus femoris<br>Rectus abdominis<br>External obliques |
| Foot turnout (eversion and external rotation) | Gastrocnemius (medial)<br>Semitendinosus<br>Semimembranosus<br>Sartorius<br>Gracilis | Soleus<br>Gastrocnemius (lateral)<br>Biceps femoris |
| Heel elevation | Tibialis (anterior) | Soleus |

| MALALIGNMENT | POSSIBLE WEAK MUSCLES | POSSIBLE TIGHT MUSCLES |
|---|---|---|
| Foot pronation (arch collapse) | Tibialis (anterior and posterior)<br>Gluteus medius<br>Gastrocnemius (medial) | Gastrocnemius (lateral)<br>Biceps femoris<br>TFL |
| Lateral weight shift | Gluteus medius (same side of shift)<br>Tibialis (anterior)<br>Adductors (opposite side of shift) | Adductors<br>TFL (same side of shift)<br>Gastrocnemius<br>Soleus<br>Piriformis<br>Gluteus medius<br>Biceps femoris (opposite side of shift) |
| Lumbar spinal extension (arching) | Gluteus maximus<br>Transverse abdominus<br>Hamstring complex<br>Multifidus | Psoas<br>Rectus femoris<br>TFL<br>Erector spinae<br>Latissimus dorsi |
| Spinal flexion (rounding) | Erector spinae | Rectus abdominus<br>External obliques<br>Pectoralis major & minor |
| Forward head | Flexors of the cervical spine<br>Extensors of the thoracic spine | Upper trapezius<br>Levator scapulae<br>sternocleidomastoid |

| MALALIGNMENT | POSSIBLE WEAK MUSCLES | POSSIBLE TIGHT MUSCLES |
|---|---|---|
| Arms fall forward | Middle and lower trapezius<br>Rhomboids<br>Supraspinatus<br>Infraspinatus<br>Teres minor<br>Subscapularis | Latissimus dorsi<br>Pectoralis major & minor<br>Teres major |
| Shoulder elevation | Middle and lower trapezius | Upper trapezius<br>Levator scapulae<br>sternocleidomastoid |

## Cardiovascular and Performance Assessments - Chapter 7

Trainers will also need a good understanding of the client's cardiovascular health before developing a program.

- Key terms
  - $VO_2$ max: maximum amount of oxygen an individual can use during exercise
    - Can be improved with training
    - An improvement in $VO_2$ max indicates higher cardiovascular fitness and performance, as well as reduced risk for heart disease and diabetes
  - Ventilatory threshold (VT): the threshold where ventilation increases faster than the volume of oxygen
  - One-repetition max (1RM): a single maximum-strength repetition with maximum load
    - Ensure the client can perform the exercise with proper form before executing a one-rep max (1RM) assessment
- Cardiovascular assessments: The general purpose of cardiovascular assessments is to determine a client's $VO_2$ max measurement. The following are a few of the most common cardiovascular assessments:
  - Cooper 12-minute run
    - The total distance completed (walk/run/combination) is recorded at the end of 12 minutes. To find the approximate $VO_2$ max, the following equations are used:
      - Using miles: (35.97 × miles completed) – 11.29 = estimated VO2max
      - Using kilometers: (22.35 × kilometers completed) – 11.29 = estimated VO2max
  - The 1.5-mile run
    - The time it takes the client to complete a 1.5-mile run; then compare the results to the norms table below:

| RATING | MALES | FEMALES |
|---|---|---|
| Very poor | >16:01 minutes | >19:01 minutes |
| Poor | 16:00–14:01 minutes | 19:00–18:31 minutes |
| Fair | 14:00–12:01 minutes | 18:30–15:55 minutes |
| Good | 12:00–10:46 minutes | 15:54–13:31 minutes |
| Excellent | 10:45–9:45 minutes | 13:30–12:30 minutes |
| Superior | <9:44 minutes | <12:29 minutes |

- Rockport one-mile walk
  - Measure body weight
  - Client walks as fast as possible, without jogging or running, for one mile
    - Males: 139.168 – (0.388 × age) – (0.077 × weight in pounds) – (3.265 × walk time in minutes) – (0.156 × heart rate) + 6.318 = estimated $VO_2$ max
    - Females: 139.168 – (0.388 × age) – (0.077 × weight in pounds) – (3.265 × walk time in minutes) – (0.156 × heart rate) = estimated $VO_2$ max
- Step test
  - Equipment needed: a step or platform that is 12 inches high (30.5 centimeters), a stopwatch, and a metronome
  - Client steps for five minutes, or until they reach exhaustion (when they cannot maintain the stepping rate for 15 consecutive seconds)
  - The trainer should ensure clients understand the tempo they are striving to keep before beginning the assessment
  - Clients should be instructed not to talk during the test and are not allowed a warm-up prior
- Strength test
  - Always consider the client's posture, range of motion, and starting position when conducting strength tests
  - A one-rep max (1RM) is an assessment typically used with intermediate to advanced clients and helps determine starting strength and intensity levels for program design
    - A bench press is the standard exercise for assessing upper body 1RM
    - A squat is the standard exercise for assessing lower body 1RM

- Muscular endurance test
  - Bench press
    - To find a safe starting point to test the client's 1RM, the trainer should choose a weight the client can lift for approximately 10 repetitions with good form
    - If the client can perform more than 20 repetitions, the trainer can allow a three- to five-minute break, increase the load by 5 to 10 percent, and conduct the test again
    - The number of completed repetitions is multiplied by 2.5; that number is subtracted from 100 and then divided by 100 to get a decimal value; this provides the estimated percentage of 1RM

| VALUES FOR BENCH PRESS STRENGTH IN 1RM/BODY WEIGHT IN POUNDS | | | | | |
|---|---|---|---|---|---|
| **Rating** | **Age in Years** | | | | |
| **Men** | **20-29** | **30-39** | **40-49** | **50-59** | **60+** |
| Excellent | >1.26 | >1.08 | >0.97 | >0.86 | >0.78 |
| Good | 1.17-1.25 | 1.01-1.07 | 0.91-0.96 | 0.81-0.85 | 0.74-0.77 |
| Average | 0.97-1.16 | 0.86-1.00 | 0.78-0.90 | 0.70-0.80 | 0.64-0.73 |
| Fair | 0.88-0.96 | 0.79-0.85 | 0.72-0.77 | 0.65-0.69 | 0.60-063 |
| Poor | <0.87 | <0.78 | <0.71 | <0.64 | <0.59 |
| **Women** | **20-29** | **30-39** | **40-49** | **50-59** | **60+** |
| Excellent | >.78 | >0.66 | >0.61 | >0.54 | >0.55 |
| Good | 0.72-0.77 | 0.62-0.65 | 0.57-0.60 | 0.53-0.59 | 0.51-0.54 |
| Average | 0.59-0.71 | 0.53-0.61 | 0.48-0.56 | 0.43-0.50 | 0.40-0.50 |
| Fair | 0.53-0.58 | 0.49-0.52 | 0.44-0.47 | 0.40-0.42 | 0.37-0.40 |

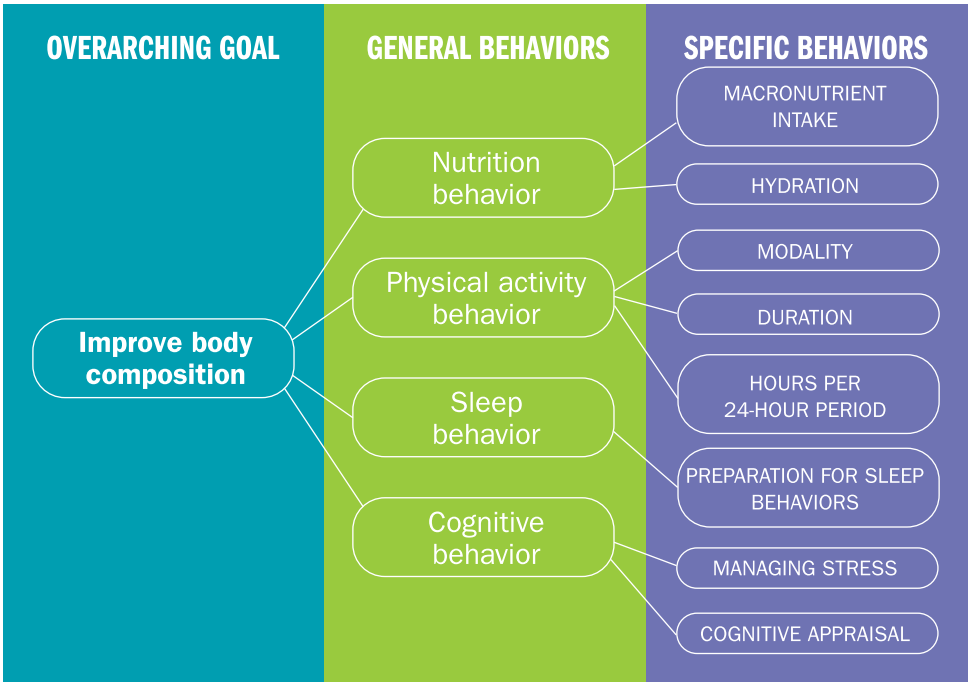
- Pushups are used for upper body muscular endurance assessments

| PUSH UP ASSESSMENT STANDARDS (NUMBER OF REPS) | | | | | |
|---|---|---|---|---|---|
| **Males** | | | | | |
| Percentile | Age | | | | |
| | **20–29 years** | **30–39 years** | **40–49 years** | **50–59 years** | **60–69 years** |
| 90 | 41 | 32 | 25 | 24 | 24 |
| 80 | 34 | 27 | 21 | 17 | 16 |
| 70 | 30 | 24 | 19 | 14 | 11 |
| 60 | 27 | 21 | 16 | 11 | 10 |
| 50 | 24 | 19 | 13 | 10 | 9 |
| 40 | 21 | 16 | 12 | 9 | 7 |
| 30 | 18 | 14 | 10 | 7 | 6 |
| 20 | 16 | 11 | 8 | 5 | 4 |
| 10 | 11 | 8 | 5 | 4 | 2 |
| **Females** | | | | | |
| Percentile | Age | | | | |
| | **20–29 years** | **30–39 years** | **40–49 years** | **50–59 years** | **60–69 years** |
| 90 | 31 | 27 | 25 | 19 | 18 |
| 80 | 27 | 22 | 21 | 17 | 15 |
| 70 | 21 | 20 | 17 | 13 | 13 |
| 60 | 19 | 17 | 16 | 12 | 11 |
| 50 | 18 | 16 | 14 | 11 | 9 |
| 40 | 14 | 13 | 11 | 9 | 6 |
| 30 | 13 | 10 | 10 | 6 | 4 |
| 20 | 10 | 7 | 8 | 3 | 0 |
| 10 | 6 | 1 | 4 | 0 | 0 |
| 90th percentile = excellent; 70th percentile = above average; 50th percentile = average; 30th percentile = below average; 10th percentile = poor | | | | | |

# Behavior Change and Motivation - Chapter 2

Arguably, the science of motivation and behavior change is the most important aspect of a successful health and wellness program. It is important for fitness professionals to have solid foundational knowledge in both to effectively help clients make lasting change.

- Behavior: an action that can be observed, measured, and modified.




- Motivation is the direction and intensity of effort
  - Intrinsic motivation: the drive to execute behaviors that are driven by internal or personal rewards
  - Extrinsic motivation: the drive to perform certain behaviors based on external factors such as praise, recognition, and money
- The transtheoretical model (TTM) is a multifaceted framework that encompasses four key distinct but interrelated constructs
  - 1. Stages of change: a person's stage of readiness to engage in the healthier/desired behavior
  - 2. Processes of change: factors influencing how a person transitions from one stage to another
  - 3. Self-efficacy: the person's belief in their capability to enact the goal behavior
  - 4. Decisional balance: determination of whether the person deems the change process worth pursuing
- Self-determination theory (SDT): a general theory of human motivation that suggests a person is motivated to change by three basic psychological needs of:
  - Autonomy: the need for self-governance and control over one's own behaviors
    - Minimize pressure
    - Avoid controlling behavior

    - Acknowledge the client's feelings
  - Competence: the basic need to feel a sense of mastery and operate effectively within the environment
    - Provide structure and routine
    - Provide the optimal level of challenge
    - Provide the opportunity to learn and master new skills
    - Provide feedback
  - Relatedness: the need to feel connected to and supported by others, as well as a sense of belonging within a group
    - Allow for meaningful interpersonal interactions
    - Promoting positive emotions to strengthen connections
    - Promote camaraderie and cohesion
    - Engage in effective communication
- To effectively help a fitness client make lifestyle changes that support their goals and promote longevity, a personal trainer should be able to:
  - Determine their client's stage of change and identify the next steps to support healthy behavior change

| STAGE OF CHANGE | DESCRIPTION | KEY INDICATORS |
|---|---|---|
| Pre-Contemplation | In denial or ignorant that a change is necessary, possible, or worth the effort within the next six months | Lacks the belief that they could change (quit a behavior or start a behavior), even if they wanted to<br>Perceives no control over their behavior (e.g., resigned to their current state)<br>Defeated by failed prior attempts to change |
| Contemplation | Contemplate making a change in the next six months but reluctant to commit | Aware of potential benefits of making a change but perceive the costs (e.g., time, effort, sacrifices) as outweighing the benefits<br>Procrastinates in making efforts toward the behavior change |
| Preparation | Committed to making a change in the target behavior within 30 days; engages in preparation activities | Begins to take small steps toward the target behavior change (e.g., buys exercise apparel or equipment, signs up for a gym membership, collects information, initiates the hire of a personal trainer) |

| STAGE OF CHANGE | DESCRIPTION | KEY INDICATORS |
|---|---|---|
| Action | Engaged in change behavior for less than six months; new behavior is not fully stabilized | Actively doing things to change or modify behavior<br><br>Structures their environment in ways that support their healthy behavior (e.g., avoids temptations that trigger the undesirable behavior) |
| Maintenance | Sustaining their new, healthy behavior for more than six months | ew behavior becomes second nature<br><br>Greater confidence in the ability to maintain the new behavior<br><br>Greater sense of control over their behavior |

- Understand the three underlying components of self-determined behavior to support program design and training session execution:
  - Autonomy
  - Competence
  - Relateness
- Use motivational interviewing: a collaborative, client-based method of guiding a client toward a self-identified motivation for change
  - Can help a personal trainer discover a client's unique motivations
  - OARS model: a communication model for motivational interviewing that includes:
    - Open-ended questions
      - Questions that require more than a yes or no answer and encourage the client to communicate the "how" and "why"
    - Affirmations
      - Affirming a personal strength or ability of the client; affirming what the client has already done or done well
    - Reflective listening
      - Listening with the intent to understand, observe client body language and behavior, and offer a reflection of what was said
    - Summarizing
      - Providing a collective summary of what was talked about, making connections between client's own responses, or summarizing the plan of action moving forward
  - Build rapport: a close, harmonious relationship in which all parties involved understand one another's feelings and communicate well

- Resist the urge to fix
- Understand the client's motivation to change
- Listen to the client
  - Have empathy: the ability to understand and share in the feelings of others
- Empower the client

- Use the tool of goal setting most effectively
  - Types of goals
    - Process goal: the focus is on the process or action that will lead to the desired end result
    - Outcome goal: the end result is a specific desired outcome
    - Subjective goal: based on an outcome that is dependent on the interpretation of the individual client
    - Objective goal: based on objective, quantifiable data that can be measured and evaluated
  - SMART goals
    - Specific: the goal is well-defined and clear as to what the client intends to do; directs specific behavior to be taken
    - Measurable: the goal provides the specific criteria to follow; it clearly identifies the extent to which the action needs to occur, such as running two miles, exercising for 36 minutes, or working out three times a week
    - Achievable: achievability depends upon the individual client, particularly the client's time constraints, physical abilities (i.e., fitness level), and mental abilities (i.e., self-efficacy)
    - Relevant: the goal demonstrates relevance because it addresses an important health behavior (e.g., exercise) for weight management; an additional check for relevance is to ensure that short-term goals align with the client's long-term/outcome goals
    - Time-bound: the goal has a clearly defined time frame (e.g., within seven days, the client plans to walk three times); having a timeline prevents procrastination and creates a sense of urgency; other goals may incorporate a deadline or date to which the goal will be achieved (e.g., "I will lose five pounds by February 1")
  - Implementation intentions: a preset plan that links critical situations (e.g., anticipated obstacles or opportunities) to goal-directed responses
  - Monitoring: the process of observing and taking notice of routine behaviors that impact goal progress and achievement
    - Self-monitoring
    - Use of technology for tracking behavior
  - Feedback
    - Provide constructive criticism

- Processes of change: the strategies and techniques that can influence an individual's transition from one stage of change to the next

| EXPERIENTIAL PROCESSES OF CHANGE | DESCRIPTION | PRACTICAL SUGGESTIONS |
|---|---|---|
| Consciousness-raising | **Getting the facts.** Increasing information about self and of the unhealthy, undesired behavior (current state of behavior) and/or their potential new behavior | Providing factual information and data relevant to the client's current behavior and target behavior |
| Dramatic relief | **Invoking emotions.** Experiencing and expressing emotional reactions to the idea of continuing the unhealthy behavior (e.g., staying the same) and to the idea of initiating a change (e.g., enacting the healthy behavior) | Using MI techniques to engage the client's emotions (e.g., engaging in supportive, empathetic listening to allow clients to express emotions without judgment or insinuating a need for them to stop or change the emotion) |
| Self-reevaluation | **Creating a new self-image.** Rethinking one's self-image to include the possibility of a successful behavior change | Helping clients clarify their core values, identifying healthy role models, and visualizing oneself reaching and maintaining the desired behavior change |
| Environmental reevaluation | **Realizing the effect on others.** Reflecting on how one's current behavior affects the physical environment and people around them to include those they care about | Asking the client to reflect on how their behaviors are impacting others around them; asking the client to consider another person's perspective on the impact of their behavior (e.g., increase empathy) |
| Social liberation | **Noticing societal acceptance.** Increasing awareness of how the healthy, desired behavior is supported by society | Helping clients realize how the desired, healthier behavior is valued within their social communities and society at large; helping bridge the client's motivational need of a sense of connectedness to others who actively engage in the target behavior, such as societal role models and/or people within the client's social circle |

| BEHAVIORAL PROCESSES OF CHANGE | DESCRIPTION | PRACTICAL SUGGESTIONS |
|---|---|---|
| Self-liberation | **Committing with confidence.** Committing to take action with the belief that change is possible | Ensuring the training program/intervention accounts for motivational needs of autonomy and competence; for example, enhancing autonomy by providing multiple choices or options within the behavior change plan, and enhancing competence by engaging in self-efficacy strategies (e.g., vicarious experience—sharing relevant examples of success and testimonies of relatable individuals) |
| Helping relationships | **Generating social support.** Establishing relationships where one feels safe to share personal challenges and receive support such as encouragement and guidance | Helping clients connect to social groups or individuals with similar goals and values, virtually and/or in-person; encouraging clients to recruit accountability partners or small support groups |
| Counterconditioning | **Making substitutions.** Finding healthier alternatives for unhealthy behaviors | Helping clients identify healthier behaviors that can be substituted for less healthy or problem behaviors |
| Reinforcement management | **Using rewards and feedback.** Using rewards and feedback strategically to reinforce positive behavior and acting on one's values | Helping clients create a plan to celebrate small and big successes, such as using rewards for accomplishing short-term goals; providing feedback and positive reinforcement when client engages in positive behaviors; teaching clients to capitalize on self-monitoring techniques (e.g., smart apps, journaling) |
| Stimulus control | **Managing the environment.** Avoiding stimuli that trigger the unhealthy behavior and intentionally creating cues that trigger the healthy behavior | Helping clients identify purposeful cues in their environment to trigger the healthy, desired behavior (e.g., creating implementation intentions); helping clients identify ways to restructure their environment to remove or overcome stimuli that trigger the unhealthy, undesired behavior |